Reconstructing arbitrary leaf multiplicities

Three quadratic moments over clique-prism cores
New notebook progress and an exact order-eight audit · 2 November 2026

Draft status: AI-assisted derivation and self-audit; independent review pending, novelty unassessed. The general graph reconstruction conjecture remains open. This report extends the earlier local notebook result; it does not claim priority over the endvertex reconstruction literature.

1. Draft theorem and exact scope

Let $H_s=K_s \square K_2$, $s\geq 3$: two s-cliques joined by a perfect matching. At each of its 2s vertices attach any nonnegative number of new leaves, directly to that vertex. Let $r\geq 1$ be the total number of added leaves and $n=2s+r$. Every card is unlabeled and repeated isomorphism types retain their multiplicities.

Theorem. The leaf-deletion deck alone determines the resulting graph up to isomorphism if $r=1$ or $r\geq 3$. If $r=2$, exactly four nonisomorphic graphs share the same leaf-deletion deck, but the full vertex-deleted deck distinguishes them. Consequently every such leaf extension is reconstructible from its full vertex deck.

leaf count	leaf-deletion deck	full vertex deck
r=1	determines the graph	determines the graph
r=2	four possible types	determines the graph
r≥3	determines the graph	determines the graph

For $r\geq 3$ a constructive algorithm recovers three quadratic moments, selects one suitable card and restores the missing leaf. Given adjacency matrices, it uses $O(n^3)$ arithmetic operations on O(log n)-bit integers. It enumerates neither graph automorphisms nor all attachment vectors. This is a class-specific algorithm, not a general reconstruction algorithm.

Delta from the preceding session

The earlier prism argument handled two leaves by path counts. The earlier arbitrary-core multiplicity rule required an absent predecessor among occupancy values. The theorem here handles every multiplicity pattern on clique-prism cores, including patterns where that gap rule fails, and needs only the leaf cards except when r=2.

2. Recognizing the core without hidden labels

Name the two cliques temporarily as upper and lower. For matching column i, let a_i,b_i be the numbers of attached leaves at its upper and lower vertices. A leaf card decrements one positive coordinate by one; a coordinate of value q produces q copies of that card. The row names and the column ordering are not part of the input.

Why this structure is visible

Core vertices have degree at least $s \geq 3$; all added vertices have degree one. Removing the degree-one vertices from any leaf card exposes H_s. Inside this core, matching edges have zero common neighbors, while a clique edge has $s-2>0$. Thus the two cliques and their matching have an intrinsic meaning. Equivalently the core complement is the connected bipartite graph $K_{s,s}$ minus a perfect matching; its two parts are unique up to exchange.

Any simultaneous permutation of matching columns, and an exchange of the two rows, is an automorphism. Conversely every core automorphism preserves the clique/matching distinction and has this form. Occupancy configurations are therefore identified exactly by a multiset of ordered column pairs (a_i,b_i), modulo transposing every pair simultaneously.

Recognition against arbitrary competing graphs

Suppose an arbitrary graph J has the same leaf-deletion deck as G. It has the same number r of leaves, since that is the number of cards in this restricted deck. A common card C has exactly $r-1$ degree-one vertices, all directly attached to the exposed core. Completing C to J adds one vertex of degree one. Attaching it to an existing leaf would turn that leaf into a degree-two vertex, leaving only $r-1$ leaves in J, a contradiction. C has no isolated vertices. Hence the new leaf attaches to the core and J has the same model.

For a full deck, recover $m=\Sigma_C|E(C)|/(n-2)$ and each deleted degree $m-|E(C)|$. This identifies the r leaf cards and reduces to the preceding argument. Thus the proof does not assume in advance that a competing full-deck reconstruction belongs to the prism class.

This colored-core viewpoint is established in the endvertex literature, including Lauri (1993). The present proof makes the recognition and unlabeled-row requirements explicit. The restriction $s \geq 3$ is necessary here: at $s=2$ the core is C_4, whose automorphisms do not intrinsically distinguish the chosen matching from the clique edges.

3. The three recoverable moments

Define $A=\Sigma a_i$, $B=\Sigma b_i$, $r=A+B$, and $D=A-B$. The following numbers are unchanged by any column permutation or row exchange:

$$Q=\Sigma_i(a_i^2+b_i^2),\ R=\Sigma_i a_i b_i,\ D^2=(A-B)^2.$$

For a leaf card C, compute Q_C, R_C, D_C^2 by the same formulas in any independently chosen labeling of its exposed core. Sums below run over the entire leaf deck with multiplicity:

$$\Sigma_C Q_C=(r-2)Q+r,$$
$$\Sigma_C R_C=(r-2)R,$$
$$\Sigma_C D_C^2=(r-2)D^2+r.$$

Proof of all three identities

Deleting an upper leaf at column i changes Q to $Q-2a_i+1$ and R to $R-b_i$. Its card occurs a_i times. A lower deletion changes them to $Q-2b_i+1$ and $R-a_i$, with multiplicity b_i. Summing gives $rQ-2Q+r$ and $rR-2R$, respectively.

An upper deletion changes D to $D-1$; a lower deletion changes it to $D+1$. Therefore the last sum is

$$A(D-1)^2+B(D+1)^2$$
$$=rD^2-2(A-B)D+r=(r-2)D^2+r.$$

For $r\geq 3$, division by $r-2$ recovers Q, R, D^2 exactly. No row or vertex alignment between different cards is used. The identities are invariant under each card independently reversing its row names.

Two local quantities from a single card

If the deleted leaf came from a vertex with original occupancy q and its matching partner has occupancy t, then

$$q=(Q-Q_C+1)/2,\ t=R-R_C.$$

The missing vertex must be restored at a core coordinate whose current occupancy is $q-1$ and whose matching partner has occupancy t. The remaining issue is to determine the row and show that multiple candidate columns are equivalent.

4. Completing one card, with all exceptions

Choose a leaf card whose two row totals are unequal, and in an arbitrary row naming write $d=A_C-B_C\neq 0$. Adding the missing leaf to the upper row would give imbalance $d+1$; adding it to the lower row would give $d-1$. Exactly one satisfies

$$(d+\varepsilon)^2=D^2,\ \varepsilon\in\{+1,-1\},$$

because the two candidate squares differ by $4d\neq 0$ and one is the actual completion. Thus the row is known. In that row, find a coordinate with occupancy $q-1$ and matching partner occupancy t. At least one exists. If there are several, exchanging their entire matching columns preserves the card and the core, since those columns have identical ordered weights. All completions are isomorphic. This recovers G.

A suitable card always exists for r≥3

If every leaf card had equal row totals, any positive upper occupancy would force $D=1$ and any positive lower occupancy would force $D=-1$. Both rows cannot then contain leaves. If only one row contains them, its total is r and every deletion balances the rows only when $r=1$. This contradicts $r\geq 3$. The card selection is therefore justified.

The small-r cases and the full-deck corollary

For $r=1$, the only leaf card is H_s, whose vertex transitivity makes all one-leaf completions isomorphic. For $r=2$ every leaf card is a one-leaf extension of H_s, so all four unordered attachment-pair types give the same two-card leaf deck. They are distinct: same base; two bases in one clique; matched bases across cliques; unmatched bases across cliques.

For the full deck, Kelly counting determines every proper path count $c(P_k,G)$. Subtract its counts in the two leaf cards and add its count in H_s. The result counts paths containing both leaves; they must be endpoints. For $k=3,4,5$ this recovers the core path signature (p_0,p_1,p_2). The four types have signatures $(1,0,0)$, $(0,1,s-2)$, $(0,1,0)$, $(0,0,2)$, all distinct. These patterns are proper because $n=2s+2\geq 8$. Thus the full deck resolves $r=2$ as well.

Algorithmic bound

For $r\geq 3$, scan each supplied adjacency matrix, strip leaves, bipartition the core complement, identify matching partners and compute weights and moments. These operations cost $O(rn^2)\leq O(n^3)$ arithmetic operations; selecting leaf cards from a full deck has the same upper order. Moment sums are $O(n^3)$, so all integers have $O(\log n)$ bits. Restore any valid candidate in one suitable card. For $r=1$ the same polynomial bound suffices. For $r=2$, direct counting of fixed paths through five vertices gives an $O(n^6)$ full-deck algorithm; no optimality is claimed.

5. Scope checks and a worked reconstruction

Take four columns with weights {(0,0),(0,1),(1,0),(1,1)}. There are four leaves, no positive occupancy with an absent predecessor, and the old elementary gap condition cannot apply. The new method succeeds. Delete an upper leaf from the (1,1) column; a canonical representation of the resulting card is {(0,0),(0,1),(0,1),(1,0)}.

quantity	original graph	selected card
Q	4	3
R	1	0
D^2	0	1
row totals	2, 2	1, 2

In this card $d=-1$, so the added leaf must go to the upper row to recover $D^2=0$. The inferred q is $(4-3+1)/2=1$ and $t=1-0=1$. Both upper candidates of type (0,1) are interchangeable columns and give the same reconstructed graph.

The three numbers alone are not a complete invariant

At $s=3$ and $r=5$ the inequivalent column multisets

$$\{(0,1),(0,1),(2,1)\} \text{ and } \{(0,1),(1,0),(1,2)\}$$

both have $(Q,R,D^2)=(7,2,1)$. They are neither equal under column permutations nor related by transposing every pair, so the graphs are nonisomorphic. Their leaf decks differ. The theorem uses the recovered moments together with an actual card; omitting that card would be a false stronger claim.

Exact corroboration

The solver takes only unlabeled leaf-card configurations, never the source weights. Exhaustive tests cover 1,884 configuration orbits for $s=3,\dots,6$ and $r=1,\dots,8$, reconstructing all 1,864 cases with $r\geq 3$ and confirming the four-way $r=2$ ambiguity. A separate extractor recovers the configurations from 40 graph decks whose card vertices were independently relabeled. Another 120 seeded cases test larger cores and weights. These are finite self-audits; the theorem rests on Sections 2–4, not on extrapolation.

6. Exact audit at core order eight

The saved proper-path certificate had passed all eligible cores through order seven. This session checked every connected minimum-degree-two core of order eight in Brendan McKay’s catalogue. The input contains 12,346 graphs; exact canonical comparison verifies 12,346 distinct codes, including 11,117 connected graphs. Of these, 7,442 meet the core condition.

SHA256(graph8.g6)
546a249902101c97d3aa590f93e533668
54bd0a6f405aa59bdb32d25c57f845a

All 201,678 unordered attachment-pair orbits pass the full proper-path signature test. Endpoint vertex-orbits plus path lengths at most three suffice in this particular finite range:

minimum path cutoff	cores separated at that cutoff
0	5497
1	1293
2	578
3	74

The cutoff zero often suffices because individual vertex-orbits already distinguish the pair. These finite observations do not imply one fixed cutoff works at every order; the earlier cycle construction disproves that shortcut.

Independent algorithms and provenance

The audit decodes and round-trips graph6 using McKay’s primary specification. Automorphism enumeration checks every permutation allowed by invariant color classes; coloring only prunes candidates and is not treated as an isomorphism test. For paths, subset dynamic programming stores the number of simple paths from a fixed source to endpoint v using exactly a vertex set S. A transition appends a neighbor outside S; summing states of each size gives the path profile.

Both new methods are cross-checked against the previous independent algorithms on all 583 eligible cores through seven vertices: full exhaustive permutation-orbit enumeration for groups, and direct depth-first path enumeration for counts. The new input hash is pinned; the verifier downloads that data only when absent. Mathematical output is deterministic; progress timing is printed separately.

This is a restricted-certificate computation, not an improvement of McKay’s established full reconstruction verification through 13 vertices. No order-eight collision was found. The signature-collision search remains open from core order nine onward, and universal separation is not proved.

7. Prior work, limits and the next claim

The main proof is a local extension of this notebook. Endvertex reconstruction, weighted/colored-core formulations and group-action tools are established literature. A bounded search did not locate an exact equivalent of this three-moment statement; that absence is not a novelty certificate. Independent review and a careful equivalence search are still required.

Josef Lauri, Endvertex-deleted Subgraphs, Ars Combinatoria 36 (1993), 171–182, Section 2, provides the relevant colored-core and group-action framework. This report does not claim to introduce that model. Primary journal PDF.

Brendan McKay's graph catalogue and format specification provide the attributed order-eight data and its encoding. Graph catalogue; format specification.

McKay, Reconstruction of small graphs and digraphs, version 4 (2022), is the existing finite benchmark. Primary paper.

Farhadian, Reconstruction of graphs via asymmetry (2016), Section 6, discusses anchor-based recovery of relative placements. Primary preprint.

The exact limitation to attack next

In the proof, equal-weight matching columns are interchangeable because every simultaneous column permutation preserves two cliques and their matching. For a noncomplete graph F in place of a clique, identical column weights need not imply a core automorphism. A useful next task is to identify sufficient symmetry for this step or give exact inequivalent completions that the proposed three-moment-plus-card method cannot separate. If the decomposition is marked, its recognizability from an unlabeled deck must be proved.

A collision of three moments, or even moments plus one card, would limit that method. It would not refute the full reconstruction conjecture unless the complete vertex decks were also rigorously equal. Arbitrary cores and leaf-free graphs remain outside this report.

Provenance: shared baseline 2026-11-02.2; result prism-arbitrary-leaf-moments and order-eight-path-certificate-audit. The same assistant performed this self-audit. Review stays needs-independent-review; novelty unassessed. Earlier attempts and proof artifacts are preserved.